

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025****Course: Second Year B. Tech****Branch Computer Engineering****Semester :IV****Subject Code & Name: BTCOC403_Y19 & Operating system****Max Marks: 60****Date: 07/07/25****Duration: 3****Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Which of the following is system software? a. Operating system b. Machine Language c. Code d. None of these		1
2	Hard real time operating system has _____ jitter than a soft real time operating system. a. less b. more c. equal d. large		1
3	Time duration required for scheduling dispatcher to stop one process and start another is known as _____ latency. a. process b. dispatch c. execution d. interrupt		1
4	A system call is a routine built into the kernel and performs a basic function. a. True b. False		1
5	The operating system is responsible for? a. bad-block recovery b. booting from disk c. disk initialization d. all of the mentioned		1
6	The operating system maintains a _____ table that keeps track of how many frames have been allocated, how many are there, and how many are available. a. memory b. mapping c. page d. frame		1
7	What are the two steps of a process execution? a. I/O & OS Burst b. CPU & I/O Burst c. Memory & I/O Burst d. OS & Memory Burs		1
8	Scheduling is done so as to _____ a. increase the waiting time b. keep the waiting time the same c. decrease the waiting time d. none of the mentioned		1

9	For a deadlock to arise, which of the following conditions must hold simultaneously?				1
	a. Mutual exclusion	b. No preemption	c. Hold and wait	d. All of the mentioned	
10	In the _____ algorithm, the disk head moves from one end to the other, servicing requests along the way. When the head reaches the other end, it immediately returns to the beginning of the disk without servicing any requests on the return trip.				1
	a. LOOK	b. SCAN	c. C-SCAN	d. C-LOOK_	
11	The _____ interrupt is used by device controllers to request service.				1
	a. nonmaskable	b. blocked	c. maskable	d. vocal	
12	There is no _____ with linked allocation.				1
	a. internal fragmentation	b. external fragmentation	c. starvation	d. all of these	
Q. 2	Solve the following.				12
A)	Explain operating services with respect to programs and users.				6
B)	Explain microkernel type operating system structure.				6
Q.3	Solve the following.				12
A)	What is inter- process communication in operating System? Explain its types.				6
B)	What is thread? Differentiates between user level thread and kernel level thread.				6
Q. 4	Solve Any Two of the following.				12
A)	Illustrate Peterson's Solution for critical section problem.				6
B)	Explain deadlock detection using resource allocation graph .				6
C)	How the readers and writers problem can be solved using semaphore?				6
Q.5	Solve Any Two of the following.				12
A)	Write short on: i. Working set model. ii. Fragmentation				6
B)	What is demand paging? Explain the steps in handling page fault using appropriate diagram.				6
C)	Assume three frames and consider the reference page string below. Reference page string: 5, 7, 6, 0, 7, 1, 7, 2, 0, 1, 7, 1, 0 Determine the number of page faults using optimal and least recently used page replacement algorithm. State which algorithm is most efficient?				6
Q. 6	Solve Any Two of the following.				12
A)	Explain the contiguous and indexed file allocation methods.				6
B)	Explain the acyclic graph directory structure.				6
C)	Explain the linked allocation type disk free space management.				6

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